Randomized Median Selection

A Sampling-Based Approach

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Outline

1. Problem and Motivation
2. Sampling Strategy and the Algorithm
3. Correctness and Linear-Time Termination
4. Failure Events and Their Probabilities
5. Takeaways

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Learning goals

- Understand the sampling-based median algorithm.
- See why one run always takes linear time.
- Prove that one run fails with probability at most $n^{-1/4}$.
- Interpret the algorithm as a Monte Carlo algorithm, and see how repetition gives a Las Vegas algorithm.

Median selection problem

Given a set S of n distinct elements from a totally ordered universe, a median is an element $m \in S$ such that

$$\#\{x \in S : x \leq m\} \geq \lceil n/2 \rceil \qquad \text{and} \qquad \#\{x \in S : x \geq m\} \geq \lceil n/2 \rceil.$$

Consider the simplifying assumptions:

- n is odd;
- all elements of S are distinct.

Hence m is exactly the $\left(\frac{n+1}{2}\right)$th element in sorted order.

Deterministic baselines

Two classical approaches

- Sort all of S: $O(n \log n)$ time.
- A deterministic $O(n)$ time selection algorithm, but more complicated. [Blum, Floyd, Pratt, Rivest, Tarjan; JCSS 1973]

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Our Objective

Design a *simpler* randomized algorithm that still runs in linear time per attempt and succeeds with high probability.

Why randomness helps here

The algorithm does *not* try to locate the exact median immediately.

Instead, it tries to find two elements $d, u \in S$ such that

1. $d \leq m \leq u$;
2. the interval
$$C := \{x \in S : d \leq x \leq u\}$$
is small enough to sort quickly.

If we can find such a narrow interval around the true median, then the remaining work is easy.

Desired sandwich interval

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- Scan S once to count $\ell_d := \#\{x \in S : x < d\}$.
- Sort only the much smaller set C.
- Output the $\left(\lceil n/2 \rceil - \ell_d + 1\right)$th smallest element of C.

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Key idea

Shrink the search space from all of S down to a small candidate interval C.

Why sorting C is sublinear?

Suppose we manage to ensure

$$|C| = o\left(\frac{n}{\log n}\right).$$

Then a standard $O(m \log m)$ sorting algorithm sorts C in time

$$O(|C| \log |C|) = o(n).$$

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Main challenge

How to find d and u quickly, without knowing the median?

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Why with replacement?

- easier to implement;
- easier to analyze because the samples are independent.

How d and u are chosen?

After sorting R, define:

$$d = \text{the } \left(\frac{1}{2}n^{3/4} - \sqrt{n}\right)\text{th smallest element of } R,$$

$$u = \text{the } \left(\frac{1}{2}n^{3/4} + \sqrt{n}\right)\text{th smallest element of } R.$$

d and u are symmetrically around the sample median, with a buffer of $\sqrt{n}$ sample positions on each side.

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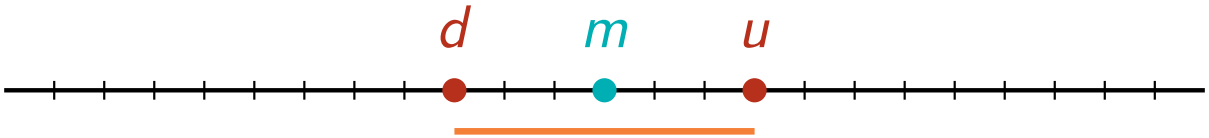
d and u are symmetrically around the sample median, with a buffer of $\sqrt{n}$ sample positions on each side.

- This buffer is what makes the true median fall inside $[d, u]$ *with high probability*.



$$C = \{x \in S : d \leq x \leq u\}$$

Goal: m lies in a "short" interval that can be sorted cheaply.

- d and u are selected from the sample R.
- C is defined in the original set S.

Algorithm: Randomized Median Algorithm (RMA) (I)

Input / output

Input: a set S of n elements over a totally ordered universe.
Output: the median element of S, denoted by m.

1. Pick a multiset R of $n^{3/4}$ elements from S, independently and uniformly at random with replacement.
2. Sort R.
3. Let d be the $\left(\frac{1}{2}n^{3/4} - \sqrt{n}\right)$th smallest element of R.
4. Let u be the $\left(\frac{1}{2}n^{3/4} + \sqrt{n}\right)$th smallest element of R.

Algorithm: Randomized Median Algorithm (RMA) (II)

5. Compare every element of S with d and u to compute
   - $C = \{x \in S : d \leq x \leq u\}$,
   - $\ell_d = \#\{x \in S : x < d\}$,
   - $\ell_u = \#\{x \in S : x > u\}$.
6. If $\ell_d > n/2$ or $\ell_u > n/2$, output FAIL.
7. If $|C| \leq 4n^{3/4}$, sort C; otherwise output FAIL.
8. Output the $\left(\lceil n/2 \rceil - \ell_d + 1\right)$th smallest element of C.

Purposes of each step

Step	Purpose
1–4	Use a random sample to guess a narrow interval around the true median.
5	Translate the sample interval $[d, u]$ back to the full set S.
6	Check whether the true median could have escaped outside $[d, u]$.
7	Guarantee the sorting work remains sublinear.
8	Recover the exact rank of the median inside C.

Numerical scale of the parameters

Take $n = 10^4$ as a sanity check.

$$n^{3/4} = 1{,}000, \qquad \sqrt{n} = 100.$$

So the algorithm:

- samples only 1,000 items from 10,000;
- chooses d and u at sample ranks $500 - 100 = 400$ and $500 + 100 = 600$;
- accepts only if $|C| \leq 4n^{3/4} = 4{,}000$.

We pay linear cost to isolate a candidate interval that is much smaller than the whole input and likely to contain the true median.

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The Main Theorem

Theorem

Algorithm RMA terminates in linear time, and if it does not output FAIL, *then it outputs the correct median element of the input set S.*

This theorem separates the analysis into two parts:

1. **deterministic correctness conditioned on no failure**;
2. **probability of failure**.

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Therefore

If $m \notin C$, step 6 forces the algorithm to output FAIL.

So if the algorithm reaches step 8, the true median must already lie in C.

Why step 8 returns the median once $m \in C$

Assume $m \in C$.

- There are exactly ℓ_d elements of S smaller than d, and all remaining elements smaller than m that are not among those ℓ_d lie inside C.
- Hence inside the sorted order of C, the median sits at position

$$\lceil n/2 \rceil - \ell_d + 1.$$

Under the event $m \in C$, step 8 returns the exact median of S.

Why one run is always linear time

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If $|C|$ is too large, step 7 stops immediately with FAIL instead of spending superlinear time.

Deterministic summary so far

What is already proved without probability?

1. Every execution halts in $O(n)$ time.
2. Any non-FAIL output is correct.

What remains?

Bound the probability that the algorithm outputs FAIL.

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Three bad events

Define

$$E_1 : \; Y_1 := \#\{r \in R : r \leq m\} < \frac{1}{2} n^{3/4} - \sqrt{n},$$

$$E_2 : \; Y_2 := \#\{r \in R : r \geq m\} < \frac{1}{2} n^{3/4} - \sqrt{n},$$

$$E_3 : \; |C| > 4n^{3/4}.$$

Intuitive meanings:

- E_1: too few sampled points land at or below the true median;
- E_2: too few sampled points land at or above the true median;
- E_3: the candidate interval in the full set is still too large.

Failure is exactly these bad events

Lemma

Algorithm RMA fails if and only if at least one of E_1, E_2, E_3 occurs.

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- Those two conditions are equivalent to E_1 and E_2, respectively.

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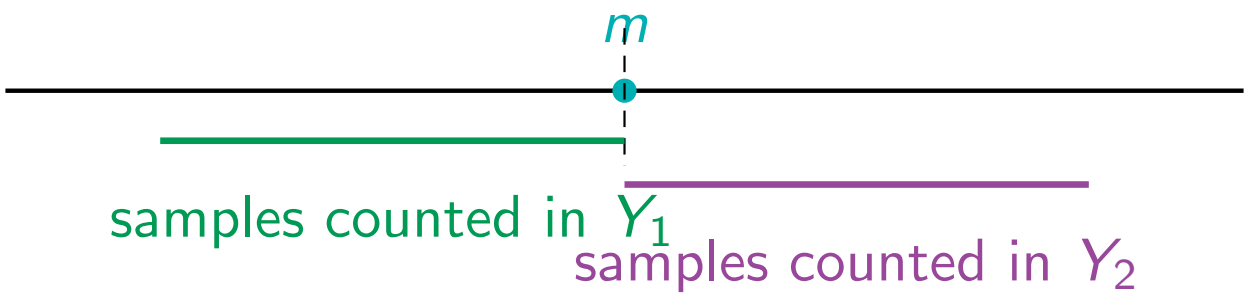
Proof Idea

Bound $\Pr[E_1]$, $\Pr[E_2]$, and $\Pr[E_3]$ separately, then apply the union bound.

Visual intuition for E_1 and E_2



If Y_1 is too small, then d ends up to the *right* of m. If Y_2 is too small, then u ends up to the *left* of m.

Bounding $\Pr[E_1]$: define indicators

For each sampled point $r_i \in R$, define

$$X_i = \begin{cases} 1, & r_i \leq m, \\ 0, & r_i > m. \end{cases}$$

Then

$$Y_1 = \sum_{i=1}^{n^{3/4}} X_i.$$

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Then

$$Y_1 = \sum_{i=1}^{n^{3/4}} X_i.$$

Because the sample is taken with replacement, $X_1, X_2, \ldots, X_{n^{3/4}}$ are independent Bernoulli trials.

Expectation of Y_1

Since m is the median and all elements are distinct,

$$\Pr[X_i = 1] = \frac{(n-1)/2 + 1}{n} = \frac{1}{2} + \frac{1}{2n}.$$

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Comparison with the bad threshold

Event E_1 asks whether Y_1 drops below

$$\frac{1}{2}n^{3/4} - \sqrt{n},$$

which is roughly $\sqrt{n}$ below its mean.

Variance of Y_1

- Because Y_1 is binomial with parameters $n^{3/4}$ and $\frac{1}{2} + \frac{1}{2n}$,

$$\mathrm{Var}[Y_1] = n^{3/4}\left(\frac{1}{2} + \frac{1}{2n}\right)\left(\frac{1}{2} - \frac{1}{2n}\right) = \frac{1}{4}n^{3/4} - \frac{1}{4n^{5/4}} < \frac{1}{4}n^{3/4}.$$

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- So the standard deviation scale is about $n^{3/8}$, while the buffer in the algorithm is $\sqrt{n} = n^{1/2}$.

Chebyshev bound for E_1

Using Chebyshev's inequality,

$$\begin{aligned}
\Pr[E_1] &= \Pr\left[Y_1 < \frac{1}{2}n^{3/4} - \sqrt{n}\right] \\
&\leq \Pr\left[|Y_1 - \mathbb{E}[Y_1]| > \sqrt{n}\right] \\
&\leq \frac{\mathrm{Var}[Y_1]}{(\sqrt{n})^2} \\
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Lemma

$$\Pr[E_1] \leq \frac{1}{4}n^{-1/4}.$$

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Cumulative progress

So far,

$$\Pr[E_1] + \Pr[E_2] \leq \frac{1}{2} n^{-1/4}.$$

Why E_3 is harder

Event E_3 is different:

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- Even if the true median lies inside $[d, u]$, the interval could still be too wide.
- We must show that the chosen sample ranks make the resulting interval in S *unlikely to contain too many elements*.

Split E_3 into two one-sided events

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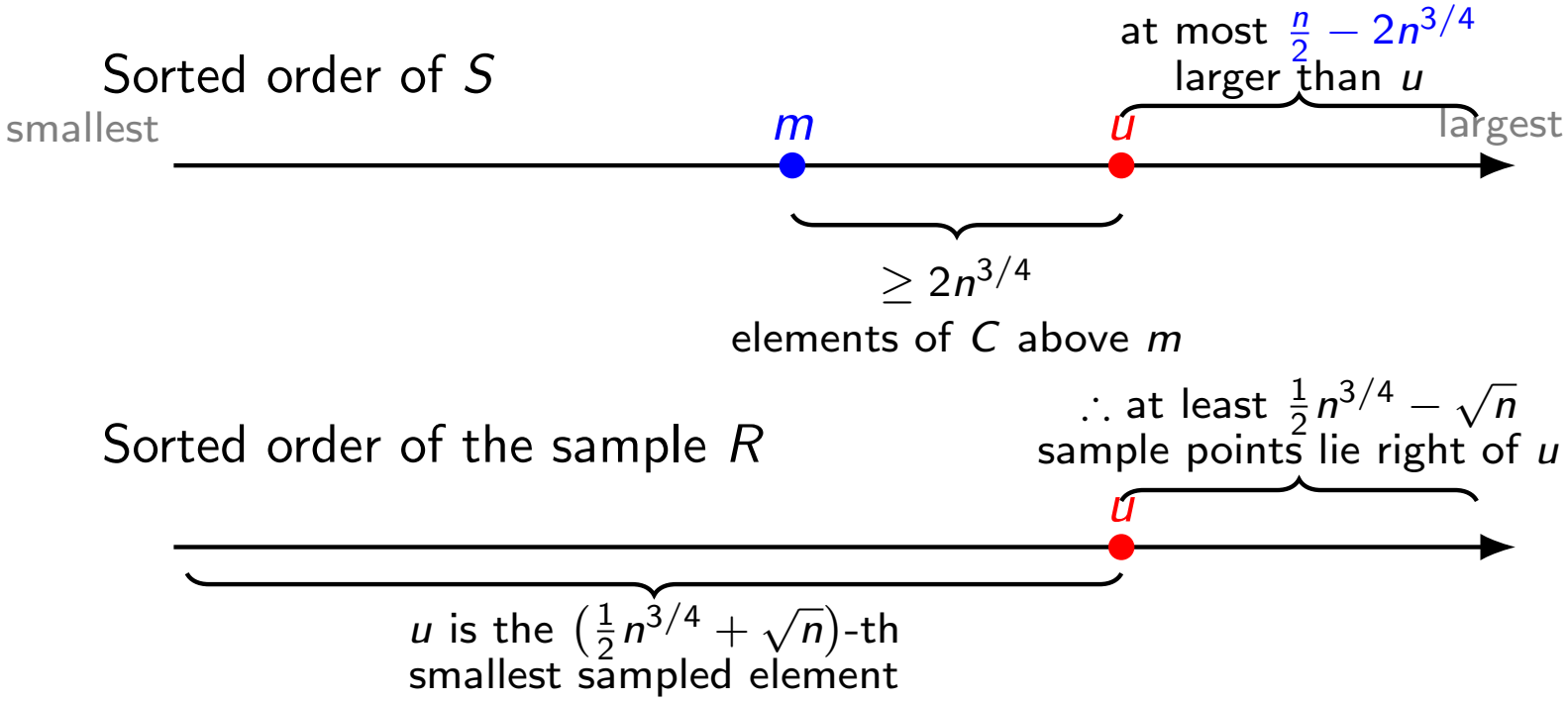
- By symmetry, it is enough to bound $\Pr[E_{3,1}]$.

What does $E_{3,1}$ imply? (1/3)

- If $E_{3,1}$ occurs, then u is too far above the median in the sorted order of S.

What does $E_{3,1}$ imply? (2/3)



$$E_{3,1} \implies \text{rank}_S(u) \geq \frac{n}{2} + 2n^{3/4}.$$

What does $E_{3,1}$ imply? (3/3)

- The rank of u in S is at least

$$\frac{1}{2}n + 2n^{3/4}.$$

- Among the sample points in R, at least $\frac{1}{2}n^{3/4} - \sqrt{n}$ samples must come from the $\frac{1}{2}n - 2n^{3/4}$ largest elements of S.

Define the variable for $E_{3,1}$

- Let X be the number of sampled points falling among the largest

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- Then $E_{3,1}$ implies $X \geq \frac{1}{2}n^{3/4} - \sqrt{n}$.

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Also,

$$\begin{aligned}\mathrm{Var}[X] &= n^{3/4}\left(\frac{1}{2} - 2n^{-1/4}\right)\left(\frac{1}{2} + 2n^{-1/4}\right) \\ &= \frac{1}{4}n^{3/4} - 4n^{1/4} < \frac{1}{4}n^{3/4}.\end{aligned}$$

Chebyshev bound for $E_{3,1}$

Now apply Chebyshev again:

$$\begin{aligned}
\Pr[E_{3,1}] &\le \Pr\left[X \ge \frac{1}{2}n^{3/4} - \sqrt{n}\right] \\
&\le \Pr\left[|X - \mathbb{E}[X]| \ge \sqrt{n}\right] \\
&\le \frac{\mathrm{Var}[X]}{n} \\
&< \frac{\frac{1}{4}n^{3/4}}{n} = \frac{1}{4}n^{-1/4}.
\end{aligned}$$

By symmetry, $\Pr[E_{3,2}] \le \frac{1}{4}n^{-1/4}$.

Chebyshev bound for $E_{3,1}$

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By symmetry, $\Pr[E_{3,2}] \le \frac{1}{4}n^{-1/4}$. Therefore,

$$\Pr[E_3] \le \frac{1}{2}n^{-1/4}.$$

Finally, using the union bound

Combining the three bounds,

$$\Pr[E_1] + \Pr[E_2] + \Pr[E_3] \leq \frac{1}{4}n^{-1/4} + \frac{1}{4}n^{-1/4} + \frac{1}{2}n^{-1/4} = n^{-1/4}.$$

Theorem

The probability that Algorithm RMA fails is at most $n^{-1/4}$.

Equivalent success guarantee

A single run succeeds with probability at least

$$1 - n^{-1/4}.$$

What was the role of Chebyshev's inequality?

For each bad event, we:

1. express a sample count as a sum of independent Bernoulli indicators;
2. compute its mean and variance;
3. use Chebyshev to control deviations of size $\sqrt{n}$.

Pedagogical point

The algorithm is a concrete application of variance calculations and Chebyshev bounds, not yet of Chernoff bounds.

Outline

1 Problem and Motivation

2 Sampling Strategy and the Algorithm

3 Correctness and Linear-Time Termination

4 Failure Events and Their Probabilities

5 Takeaways

One-run guarantee

For one execution:

- running time is always $O(n)$;
- output is either the correct median or FAIL;
- failure probability is at most $n^{-1/4}$.

This is a *Monte Carlo* algorithm: bounded running time, but a small probability of failure.

From Monte Carlo to Las Vegas

Repeat Algorithm RMA until it succeeds.

Because the samples used in different runs are independent,

- each run succeeds with probability at least $1 - n^{-1/4}$;
- the number of runs until success is geometric.

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The repeated version always returns the correct median.

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Because the samples used in different runs are independent,

- each run succeeds with probability at least $1 - n^{-1/4}$;
- the number of runs until success is geometric.

Result

The repeated version always returns the correct median.

So repetition converts the Monte Carlo algorithm into a *Las Vegas* algorithm.

Expected running time after repetition

Let $p \geq 1 - n^{-1/4}$ be the success probability of one run. For a geometric random variable with success probability p,

$$\mathbb{E}[\text{number of runs}] = \frac{1}{p}.$$

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Conclusion

The repeated algorithm is Las Vegas with linear expected running time.

Why these parameters are balanced

There are three competing design goals:

1. sample small enough so sorting R is cheap;
2. buffer large enough so m lies between d and u with high probability;
3. candidate set C small enough so sorting C is cheap.

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There are three competing design goals:

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The algorithm chooses

$$|R| = n^{3/4}, \qquad \text{buffer} = \sqrt{n}, \qquad \text{accept if } |C| \leq 4n^{3/4},$$

which makes all three constraints fit together cleanly with Chebyshev bounds.

Big-picture summary

1. Sample $n^{3/4}$ elements.
2. Use the sample to define a narrow interval $[d, u]$ around the sample median.
3. Scan the original input to form the candidate set C.
4. Abort if C is too large or if the median appears to fall outside the interval.
5. Otherwise sort only C and recover the exact median.

Main theorem chain

One run takes $O(n)$ and satisfies $\Pr[\textsc{Fail}] \leq n^{-1/4}$. Hence repeated independent restarts give a Las Vegas algorithm with linear expected running time.

Takeaways

- Guess a structure cheaply;
- Verify it deterministically;
- Retry if needed.

Discussions